

# LLMs for research: quality, speed and cost

Comparison with Pi, replaying project tickets from Imperial Dragon Harness workflows

Ten tickets randomly sampled from the author's project history.

Local and hosted models replay the same tasks; three judges score the deliverables out of 30.

We compare quality, elapsed time and cost, including incidents and retries.

## **Luna 6 medium: the best observed trade-off**

For both Claude and ChatGPT, moving up a model tier while reducing effort offers no clear advantage here: Opus 5.5 low versus Sonnet 5.5 medium, and Sol 6.1 low versus Luna 6 medium, have similar mean quality at higher cost. Sonnet is also faster than Opus; Luna has comparable speed to Sol. Haiku 5.5: similar means to Sonnet (21.70 vs 21.35/30), USD 0.13 vs 0.30, time  $\times 2.92$ ; equivalence not established. Limitation: one model fills all three roles, which may disadvantage a model built for one.

## **Qwen 3.8 Flash-Next xhigh: a local option when quality comes first**

With Strata and IQ3\_S quantization, this setup has the highest mean score: 26.2/30, taking an average of 46 minutes per ticket. It is better suited to asynchronous work. Its lead is sample-sensitive: excluding ticket 0333 brings its mean down to Mistral's.

## **Local Qwen 3.8: a high mean score against hosted references**

The best local Qwen 3.8 setup has a higher mean score than hosted references and Qwen 3.6 in this sample. The gain combines generation, quantization, runtime and effort. Qwen runs at xhigh versus medium for the references: more verbose output and longer elapsed time.

## **Latency and shared-GPU serialization limit local interactive use**

On the test workstation, latency and the need to serialize tasks on shared GPUs severely limit interactive use, while APIs allow highly parallel workflows. Future local models potentially derived from Qwen 4 and Mistral 4, hardware upgrades and daily runtime improvements may ease these constraints.

## **Mistral Large 4: promising quality, costly integration and retries**

Mistral Large 4, announced on October 6, 2026, shows promising quality. However, hosting incidents and loops observed in its integration with Pi increase elapsed time and cost for this setup; model reliability is not isolated.

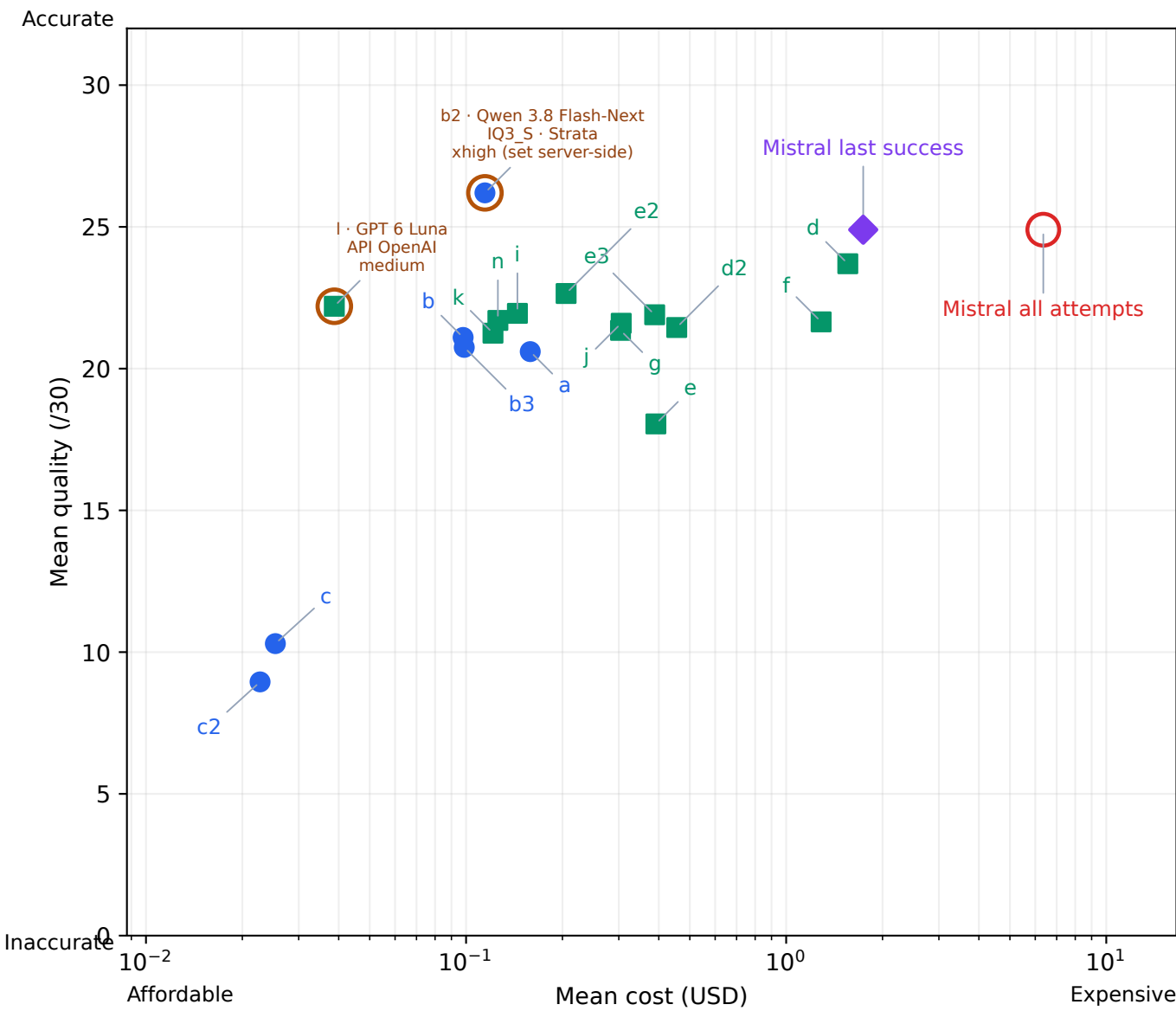
# Results by model

| Model · variant / engine · effort                                                  | Attempted | Completed / 10 | Mean quality /30 | Mean time (min / ticket) | Mean cost (USD / ticket) | Mean generated tokens / ticket |
|------------------------------------------------------------------------------------|-----------|----------------|------------------|--------------------------|--------------------------|--------------------------------|
| a · Qwen 3.8 27B<br>Q4_K_M · llama.cpp · xhigh (set server-side)                   | 16        | 9 / 10         | 20.60            | 63.87                    | 0.16                     | 68 016                         |
| b · Qwen 3.8 Flash-Next Coder<br>IQ1_M · Strata · pruned · xhigh (set server-side) | 14        | 10 / 10        | 21.10            | 39.41                    | 0.10                     | 64 702                         |
| b2 · Qwen 3.8 Flash-Next<br>IQ3_S · Strata · xhigh (set server-side)               | 11        | 10 / 10        | 26.20            | 46.11                    | 0.11                     | 71 072                         |
| b3 · Qwen 3.8 Flash-Next<br>Q2_0 · Strata · xhigh (set server-side)                | 10        | 9 / 10         | 20.75            | 39.73                    | 0.10                     | 71 416                         |
| c · Qwen 3.6 35B-A3B<br>UD-Q4_K_XL · llama.cpp · effort not measured               | 10        | 10 / 10        | 10.30            | 10.21                    | 0.03                     | 24 584                         |
| c2 · Qwen 3 Coder 30B-A3B<br>UD-Q4_K_XL · llama.cpp · effort not measured          | 10        | 9 / 10         | 8.95             | 9.15                     | 0.02                     | 11 477                         |
| d · Claude Opus 5.5<br>API Anthropic · medium                                      | 12        | 10 / 10        | 23.70            | 9.07                     | 1.56                     | 24 773                         |
| d2 · Claude Opus 5.5<br>API Anthropic · low                                        | 10        | 10 / 10        | 21.45            | 3.03                     | 0.46                     | 8 666                          |
| e · GPT 6 Sol<br>API OpenAI · medium                                               | 10        | 10 / 10        | 18.05            | 4.91                     | 0.39                     | 7 548                          |
| e2 · GPT 6.1 Sol<br>API OpenAI · low                                               | 10        | 10 / 10        | 22.65            | 5.15                     | 0.21                     | 4 687                          |
| e3 · GPT 6.1 Sol<br>API OpenAI · medium                                            | 10        | 10 / 10        | 21.90            | 8.38                     | 0.39                     | 9 425                          |
| f · GLM 5.3<br>API Mistral · high                                                  | 10        | 10 / 10        | 21.65            | 10.58                    | 1.29                     | 46 748                         |
| g · Claude Sonnet 5.5<br>API Anthropic · medium                                    | 10        | 10 / 10        | 21.35            | 2.09                     | 0.30                     | 8 896                          |
| i · DeepSeek Flash<br>API DeepSeek · high                                          | 10        | 10 / 10        | 21.95            | 9.58                     | 0.14                     | 63 848                         |
| j · GLM 5.3 Flash<br>API OpenRouter · high                                         | 15        | 9 / 10         | 21.60            | 26.96                    | 0.31                     | 46 654                         |
| k · Mimo v2.6 Flash<br>API OpenRouter · medium                                     | 13        | 9 / 10         | 21.25            | 60.06                    | 0.12                     | 125 159                        |
| l · GPT 6 Luna<br>API OpenAI · medium                                              | 10        | 10 / 10        | 22.20            | 5.02                     | 0.04                     | 16 052                         |
| mL · Mistral Large 4 · Last success<br>API Mistral + OR (0874) · off               | 35        | 10 / 10        | 24.90            | 30.60                    | 1.74                     | 59 605                         |
| mA · Mistral Large 4 · All attempts<br>API Mistral + OR (0874) · off               | 35        | 10 / 10        | 24.90            | 96.32                    | 6.36                     | 196 233                        |
| n · Claude Haiku 5.5<br>API Anthropic · medium                                     | 10        | 10 / 10        | 21.70            | 6.11                     | 0.13                     | 26 511                         |

Means per ticket, including failures; incomplete series cover evaluated tickets only. Local cost: electricity only. Mistral last success: ten successes, including 0874 via OpenRouter. Mistral all attempts: the same scores, with costs and times from 25 retained direct attempts plus the OR replacement for 0874; eight other OR attempts and one direct configuration incident are excluded. “Completed” counts delivered tickets, not attempts. The 9/10 rows reflect four two-hour timeouts (a and b3: 0874; j: 0333; k: 0211) and one termination without a valid deliverable (c2: 0470). These failures retain a zero score and consumed resources rather than being replaced by a successful retry; Mistral’s different treatment is explained in the limitations.

# Quality / cost

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



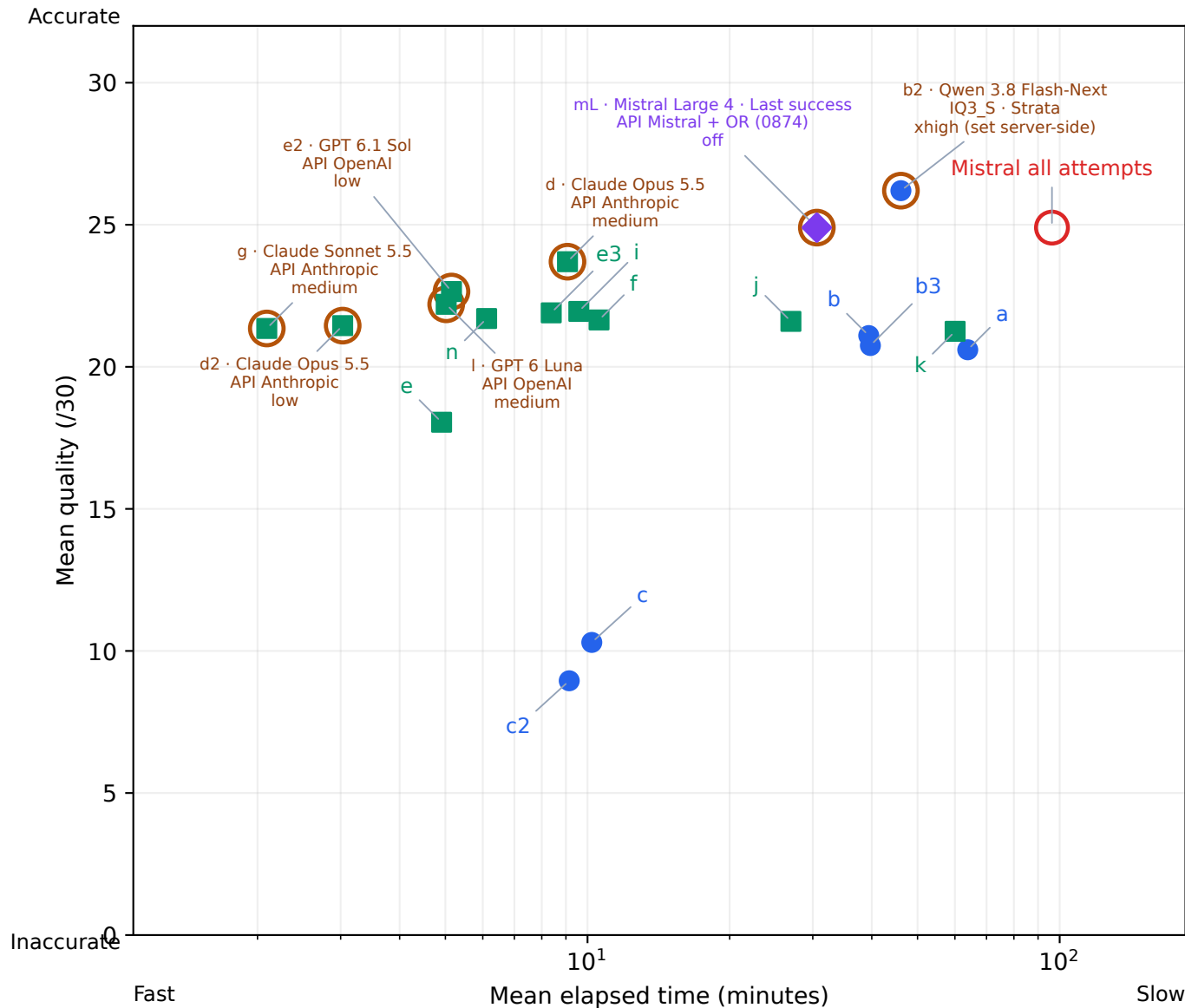
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Blue: local; green: hosted. Mistral: purple diamond = last success, red circle = all attempts. Cost and time use logarithmic scales. Orange ring: Pareto frontier (complete series, mean quality ≥ 15/30). Local cost = electricity only; hosted cost = API.

# Quality / speed

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



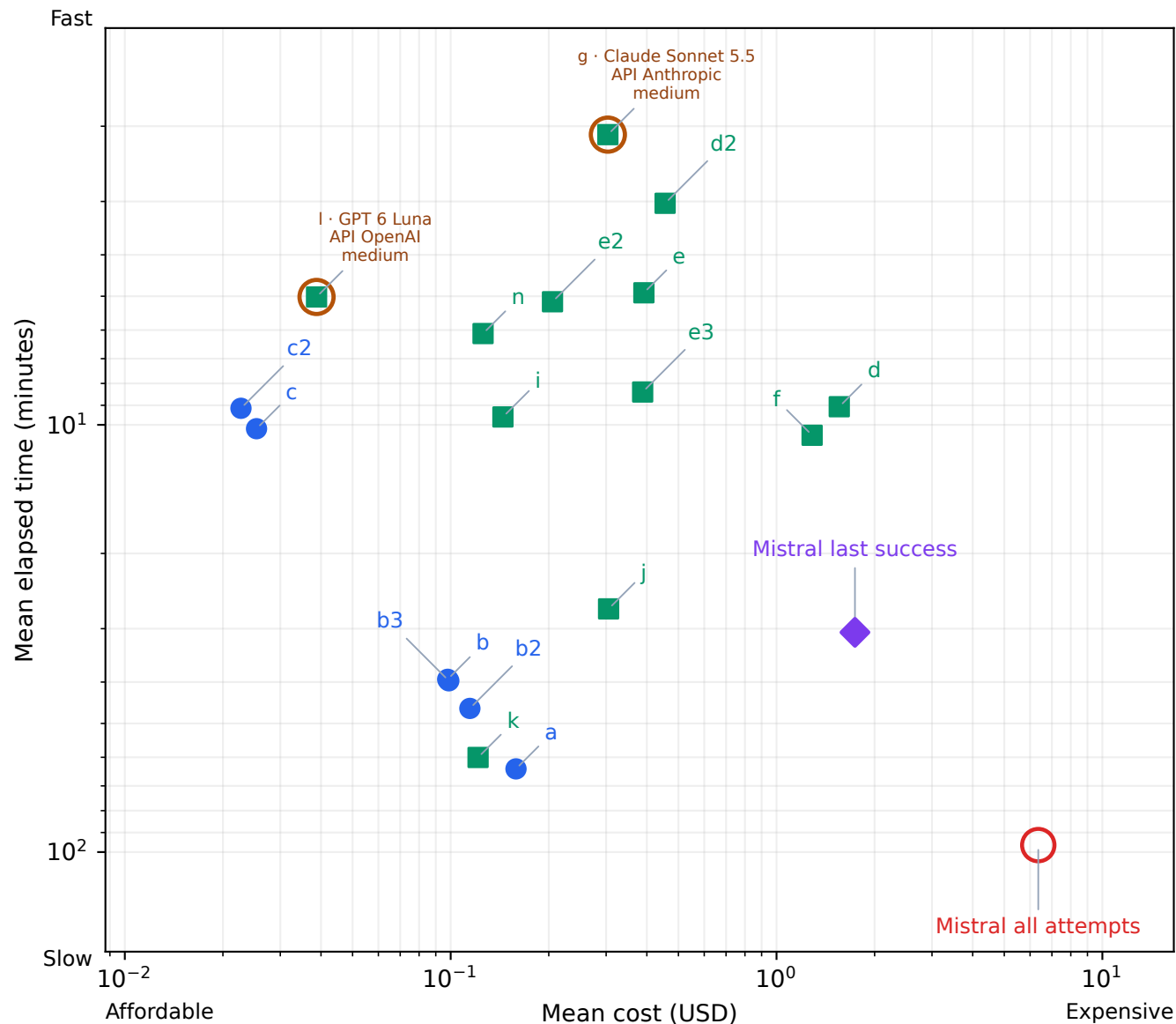
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# Speed / cost

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures



## Model · variant / engine · effort

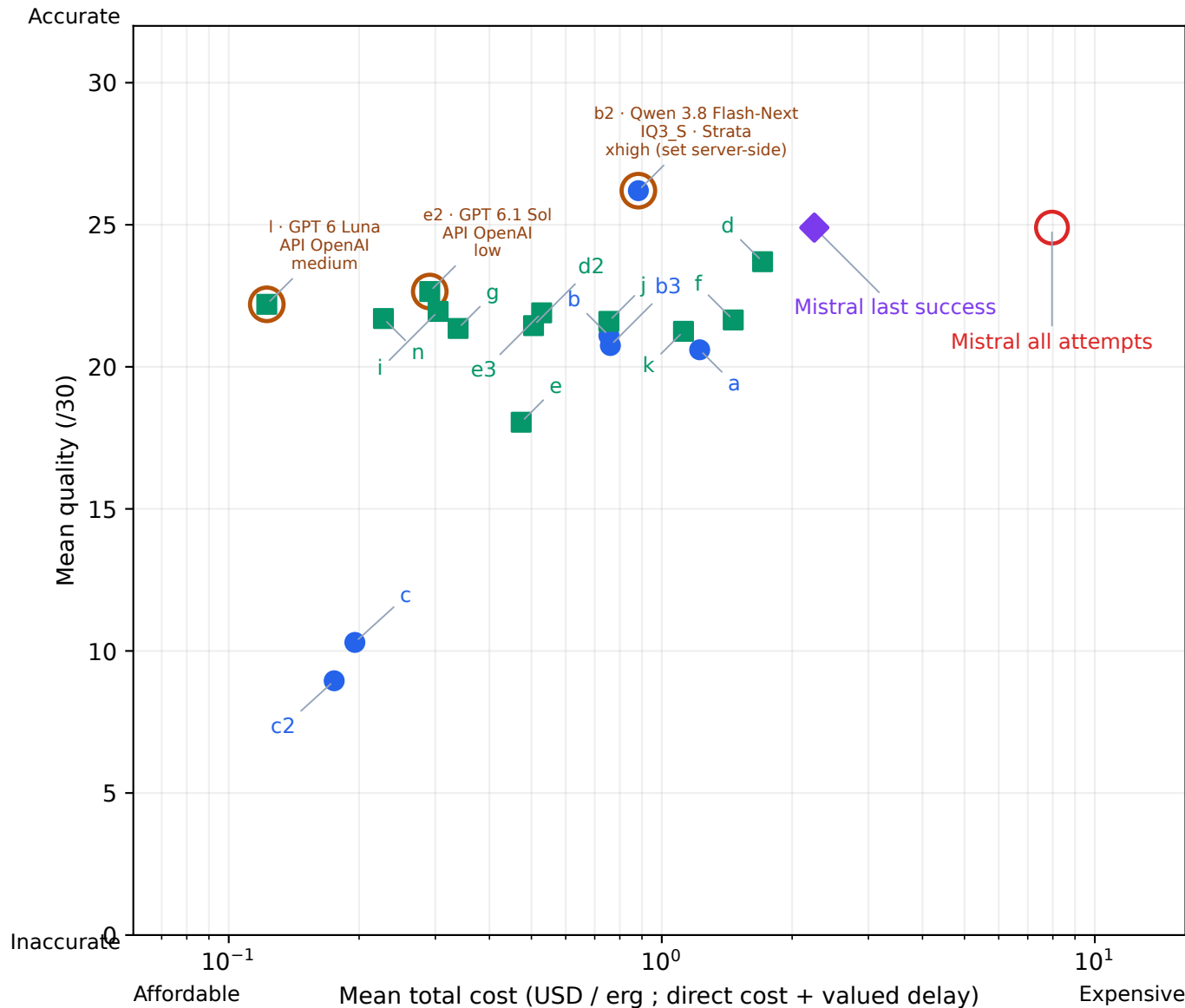
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# Quality / total cost — delay valued at 1 USD/(erg·h)

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures

Per average erg: total cost = direct cost + 1 × time / 3,600; then averaged. Time in seconds, DNF included.



## Model · variant / engine · effort

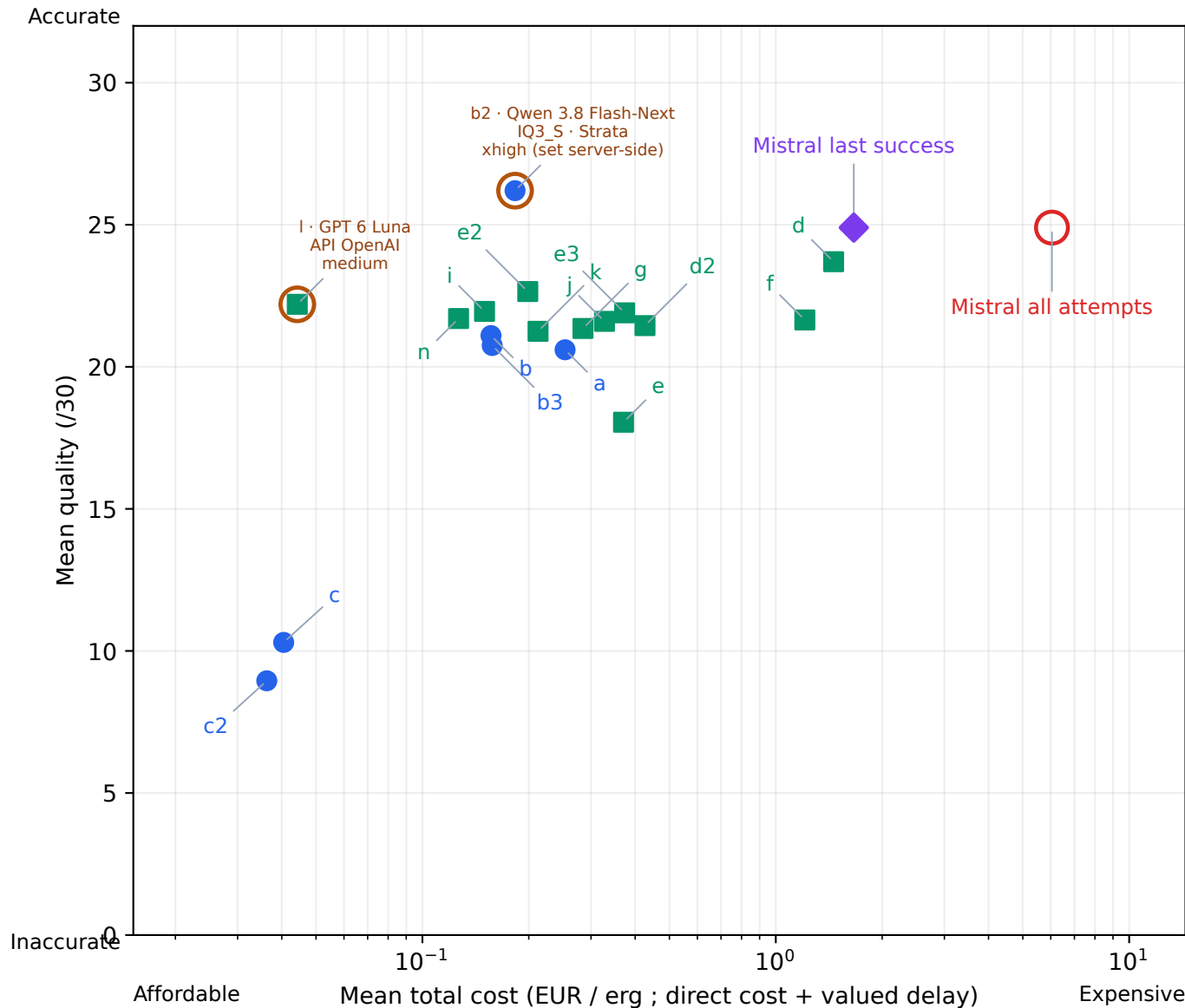
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# Quality / total cost — delay valued at 0.10 EUR/(erg·h)

Model × variant × effort · mean quality, cost and elapsed time per ticket, including failures

Per average erg: direct cost in USD / 1.08 + 0.10 × time / 3,600 (seconds), then averaged. DNF included.



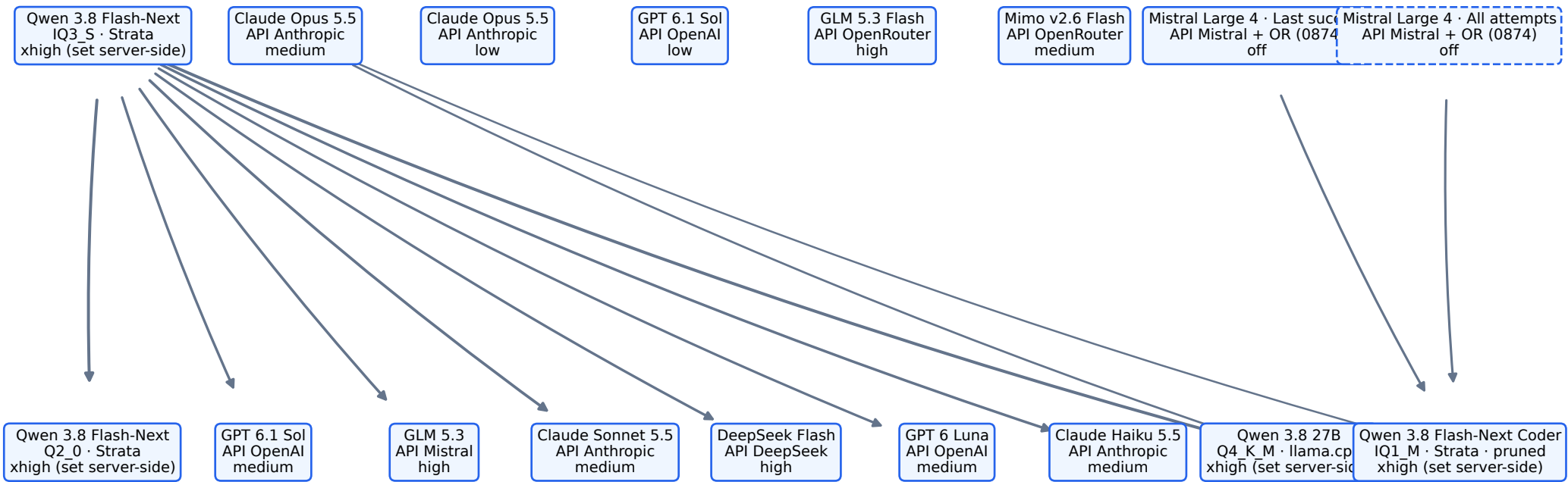
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# Quality — who beats whom?

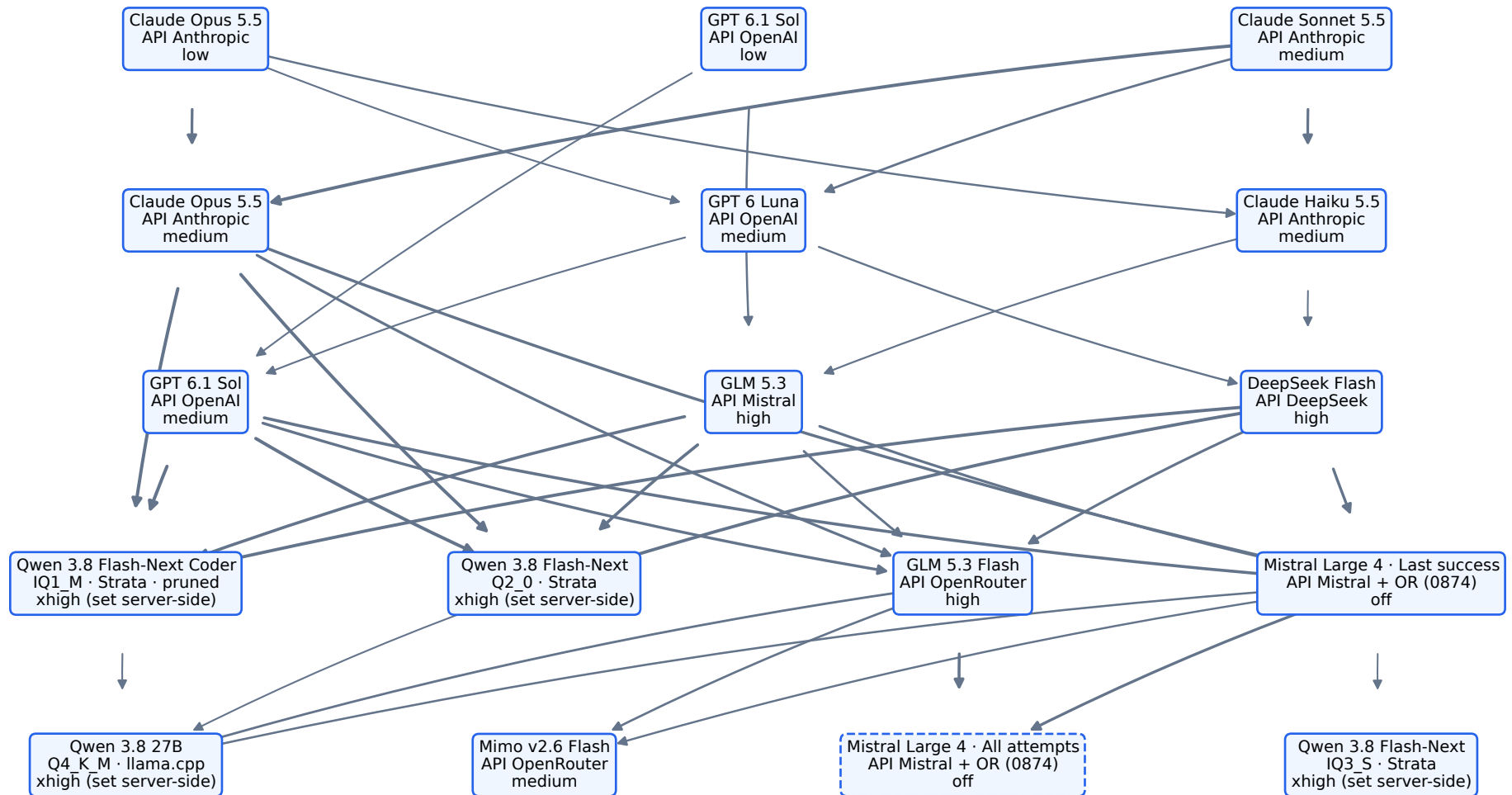
Accurate at the top · arrow towards Inaccurate · exact paired Wilcoxon · two-sided  $p < 0.05$



Arrows use the nominal 5% threshold; no comparison survives Holm correction (136 tests per axis).  
Arrow thickness indicates effect intensity. No arrow  $\neq$  equivalence.  
An indirect path is not another significant test.

# Speed — who beats whom?

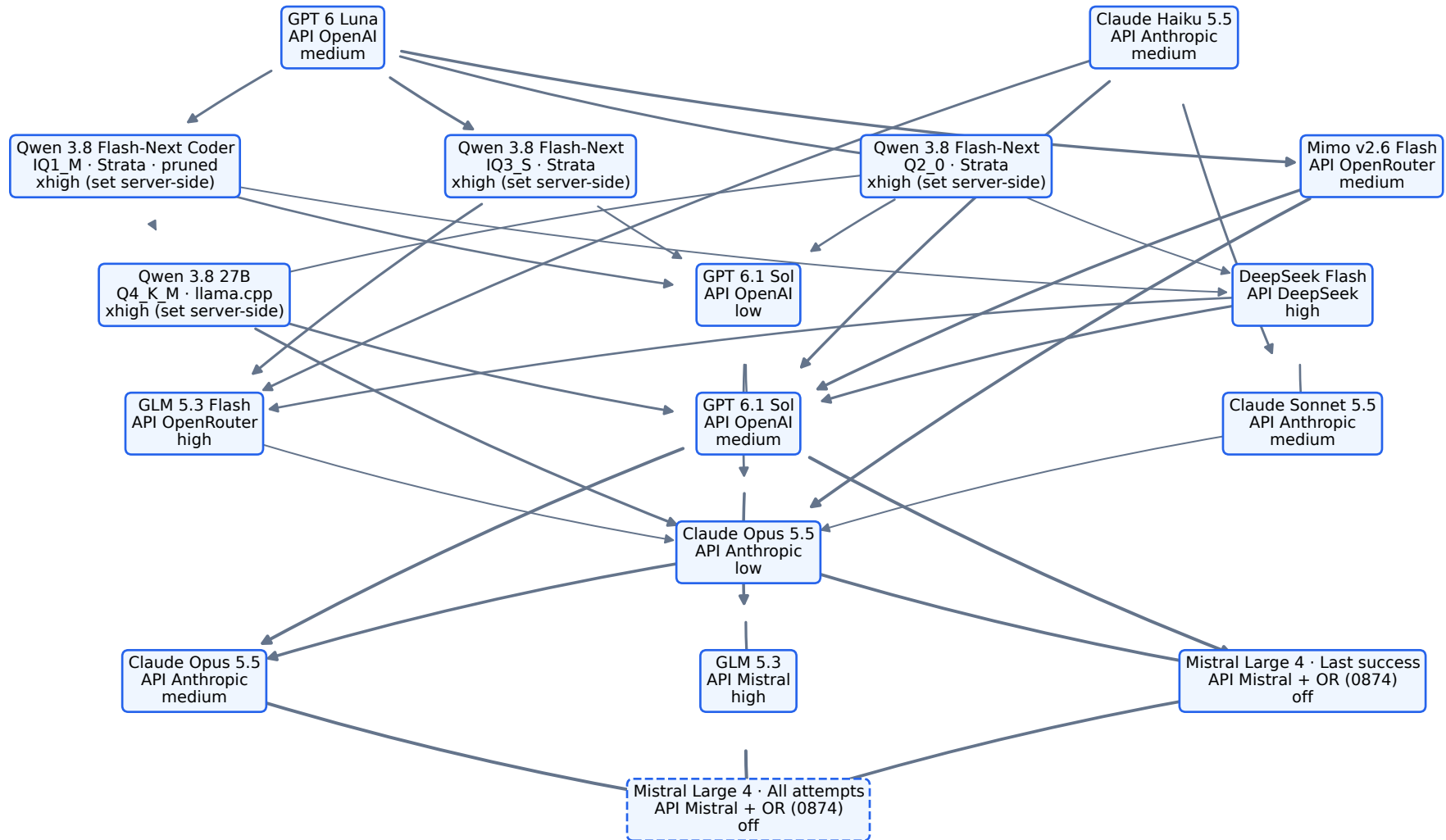
Fast at the top · arrow towards Slow · exact paired Wilcoxon · two-sided  $p < 0.05$



Arrows use the nominal 5% threshold; no comparison survives Holm correction (136 tests per axis).  
Arrow thickness indicates effect intensity. No arrow  $\neq$  equivalence.  
An indirect path is not another significant test.

# Cost — who beats whom?

Affordable at the top · arrow towards Expensive · exact paired Wilcoxon · two-sided  $p < 0.05$



Arrows use the nominal 5% threshold; no comparison survives Holm correction (136 tests per axis).  
 Arrow thickness indicates effect intensity. No arrow  $\neq$  equivalence.  
 An indirect path is not another significant test.

# Method and scope

Random sample of ten tickets from the author's project history. Each model replays the same tasks from their pre-resolution state, with the project instructions and a two-hour limit per attempt.

Pi provides the agent runtime: model access, tool calls and session management. Tickets come from projects using Imperial Dragon Harness (IDH). The prompt requires the historical instructions of each repository (AGENTS.md, tickets/AGENTS.md and associated rules). To evaluate each model separately, the tournament disables external reviews and the Gaze, review-pr and verify-gate panels: each model performs the task, runs mechanical checks and reviews its own work. The three judges intervene only after completion. This evaluates Pi with project instructions, rather than the full IDH workflow.

Boxing-style scoring out of 30: three judges start at ten and deduct points for faults according to severity:  $-3$ ,  $-2$ ,  $-1$  or  $-0.5$ . Judges inspect the work without knowing which model produced it. No usable result scores zero; a correctly explained inability to complete is judged normally.

The head judge is Gemini 3.1 Pro Preview; the other two are Grok 4.3 and MiniMax M2.5. Retained first-cycle results were judged by Gemini 3.1 Pro Preview, DeepSeek V4 Pro and Kimi K3. Scores are summed without judge weighting.

Local models run on a Lenovo ThinkStation P620: Ryzen Threadripper PRO 3945WX, 128 GB RAM, RTX A4000 + RTX 3060 GPUs (28 GB VRAM total). Local cost estimates electricity at 600 W and EUR 0.23/kWh, without hardware depreciation. API costs cover consumed tokens; the common conversion is USD 1.08 per EUR 1.

Scatterplots show mean quality, elapsed time and cost per ticket. Each ticket represents one average erg (unit of completed work). Delay is valued per erg per hour, in USD/(erg·h) or EUR/(erg·h), rather than as an hourly wage. The relevant cost is that of the delivered result: retries consume resources and delay delivery. Mistral last success retains the latest successful result per ticket, which is also the best on the three tickets with retries. Mistral all attempts sums the costs and times of 25 retained direct attempts and the OpenRouter success on 0874, including interruptions, and retains the best score. Eight other OpenRouter attempts and one direct configuration-invalidated attempt are excluded. Costs are allocated by billed tokens; newer attempts remain estimates. Pareto eligibility requires complete series and mean quality of at least 15/30.

A May 2026 benchmark of Vietnam's thermal power inventory, with five repetitions per condition, showed the value of source documents and commercial models' repeatability. Here we assess work delivered with tools rather than the factual accuracy of an inventory.

- Pi : [github.com/earendil-works/pi](https://github.com/earendil-works/pi)
- IDH : [github.com/MinhHaDuong/ImperialDragonHarness](https://github.com/MinhHaDuong/ImperialDragonHarness)
- Comparison data: frozen snapshot of October 8, 2026
- May 2026 benchmark: Beyond RAG (slides)

# Method: comparisons and limitations

In the three directed acyclic graphs, arrows compare paired ticket results with a two-sided Wilcoxon signed-rank test at the 5% level. They point towards less accurate, slower or more expensive results. Transitively redundant arrows are removed for readability. No arrow does not establish equivalence; an indirect path is not an additional test.

The 5% threshold applies to each comparison, without a simultaneous guarantee for the whole graph. Holm correction controls the false-positive risk from multiple comparisons. With ten tickets and 136 comparisons per graph, the corrected threshold is unattainable for this test, even when a model wins all ten tickets. No arrows after correction therefore does not mean the models are equivalent: this sample cannot establish differences with that global guarantee. Mistral all attempts has a dashed outline in the DAGs. Qwen 3, Qwen 3.6 and GPT-6 Sol appear only in scatterplots.

Limitation: a single model performs all roles: orchestration, implementation and review of its own output. This measures its ability to complete a task end to end, rather than the performance of a harness routing roles to different models and assigning reviews to other models to reduce correlated errors. This design may disadvantage a worker model such as Haiku 5.5, or any model stronger in one role than in the others; the test does not measure that. The three external judges evaluate the final result without taking part in it.

Limitation: on ticket 0874, the last direct Mistral attempt reached the two-hour cap despite 195 successful API responses. The model spent a long time exploring the project and repeatedly testing, with missing modules and a missing hook in the environment. The identical-command repetition breaker does not catch this drift across varying commands. The successful OpenRouter result replaces this ticket in the last-success series; Mistral all attempts retains failed direct-attempt costs and times and adds the replacement. Mistral therefore combines two hosting routes. OpenRouter cost is the recorded cost of that run; direct costs follow the Mistral invoice calibration.

Limitation: ordinary series retain results after retries, whereas Mistral all attempts aggregates attempts. Repeatability means low dispersion across repetitions; first-attempt completion is another form of reliability. Heterogeneous retries prevent a uniform comparison of the latter here. “Attempted” counts retained attempts, including those invalidated by infrastructure or an incorrect starting state; these incidents are not all attributable to the model. Both Mistral rows count both providers, while all-attempts costs follow the documented scope. Mean generated tokens follow the runs used in each series, including reasoning when reported as output tokens; tokenizers differ across providers.

b2 retry: archives contain eleven attempts for ten tickets, including a 0333 attempt interrupted by SIGTERM after 17 min 24 s (exit code 143). The author confirmed requesting this pause to free the GPU for other work; the interruption was not based on output quality. The retained replay scored 30/30; the interrupted attempt had no score allowing best-score selection. This replacement and its excluded resource use mean the series is not ten first-attempt successes. Excluding 0333 from every series gives Qwen b2 and both Mistral series the same mean quality, 25.78/30 over nine tickets: b2 no longer has the sole highest mean. “Attempted” counts retained traces, not a guaranteed exhaustive history.

Candidates were selected ad hoc, without a claim to exhaustiveness. Which models, variants or runtimes would you like to see in the next comparison? Suggestions and feedback are welcome at [minh.ha-duong@cnsr.fr](mailto:minh.ha-duong@cnsr.fr).

# **Auto-generated technical appendix**

# Supplementary analyses and situation guide

October 8 frozen results; supplement prepared October 9. Guide reviewed by a delegated Fable reviewer at the author's request.

## Executive summary

H1 does not hold: Sol low is not faster than Sonnet medium. H2 is mixed: Luna has lower mean time than all six local setups, but equal quality is not established and IQ3\_S retains a higher mean score. Effort comparisons do not establish quality dominance. Routing groups are operational choices, not fitted statistical clusters.

## Contents of the completed report

Pages 1-12: summary, results, five scatterplots, three DAGs, methods and limits. Page 13: technical appendix title. Page 15: grid, medians and 3D Pareto. Pages 16-17: paired verdicts. Pages 18-22: distributions for thirteen comparisons. Page 23: routing classes. Page 24: historical outcomes and corrections. Pages 25-26: situation guide, quotas and pricing. Page 27: provenance and ticket coverage.

## Publication status

Prepared by ChatGPT, then finalized by Claude Opus 5.5, at the request of Ha-Duong Minh. A review branch is authorized; no merge, blog publication or Tchap/Reddit distribution is authorized. Message: no ranking survives Holm correction; choose by constraints first (privacy, latency, budget, GPU, quotas), then by the observed trade-off. The following guide is not an applied routing decision.

# Grid and three-dimensional Pareto frontier

Marginal medians do not replace paired differences. Each identity includes model, effort and runtime.

| Arm | Setup (effort: legends)        | Q med. | Q mean | Min med. | USD med. |
|-----|--------------------------------|--------|--------|----------|----------|
| a   | Qwen 3.8 27B                   | 22.75  | 20.60  | 51.97    | 0.1291   |
| b   | Qwen 3.8 Flash-Next Coder      | 21.25  | 21.10  | 31.30    | 0.0778   |
| b2  | Qwen 3.8 Flash-Next            | 26.50  | 26.20  | 42.87    | 0.1065   |
| b3  | Qwen 3.8 Flash-Next            | 24.00  | 20.75  | 33.13    | 0.0823   |
| c   | Qwen 3.6 35B-A3B               | 9.25   | 10.30  | 6.52     | 0.0162   |
| c2  | Qwen 3 Coder 30B-A3B           | 8.50   | 8.95   | 5.62     | 0.0140   |
| d   | Claude Opus 5.5                | 26.50  | 23.70  | 5.99     | 0.8003   |
| d2  | Claude Opus 5.5                | 22.75  | 21.45  | 2.82     | 0.3066   |
| e   | GPT 6 Sol                      | 16.00  | 18.05  | 3.53     | 0.3731   |
| e2  | GPT 6.1 Sol                    | 23.25  | 22.65  | 5.10     | 0.2066   |
| e3  | GPT 6.1 Sol                    | 25.00  | 21.90  | 9.40     | 0.3349   |
| f   | GLM 5.3                        | 22.00  | 21.65  | 8.71     | 0.5242   |
| g   | Claude Sonnet 5.5              | 21.00  | 21.35  | 1.13     | 0.1544   |
| i   | DeepSeek Flash                 | 20.50  | 21.95  | 10.15    | 0.1059   |
| j   | GLM 5.3 Flash                  | 26.00  | 21.60  | 10.39    | 0.1694   |
| k   | Mimo v2.6 Flash                | 23.00  | 21.25  | 44.89    | 0.0808   |
| l   | GPT 6 Luna                     | 21.75  | 22.20  | 3.84     | 0.0278   |
| n   | Claude Haiku 5.5               | 22.50  | 21.70  | 3.03     | 0.0227   |
| mi  | Mistral Large 4 · Last success | 25.75  | 24.90  | 30.46    | 1.2062   |
| mr  | Mistral Large 4 · All attempts | 25.75  | 24.90  | 77.02    | 2.1511   |

3D Pareto on means, complete series with quality  $\geq 15/30$ : b2, d, d2, e2, g, l, mi. A series is dominated when another has at least its quality and no higher time or cost, with one strict improvement. This is a descriptive frontier, without statistical guarantee.

# Paired verdicts: H1 and H2

Read A / B: quality wins, ties, losses; median  $\Delta Q$ ; median A/B ratios. Ratio  $<1$  means fewer resources.

| Question: A/B | W/T/L  | $\Delta Q$ | Time | Cost | p Q    | Holm Q | Dom. /10 |
|---------------|--------|------------|------|------|--------|--------|----------|
| H1 e2/g       | 4/1/5  | -0.25      | 3.53 | 0.97 | 0.5703 | 1.0000 | 0        |
| H2-a l/a      | 5/0/5  | -0.00      | 0.08 | 0.21 | 0.9805 | 1.0000 | 5        |
| H2-b l/b      | 4/0/6  | -1.25      | 0.13 | 0.27 | 0.8418 | 1.0000 | 4        |
| H2-b2 l/b2    | 1/2/7  | -2.75      | 0.11 | 0.25 | 0.0156 | 1.0000 | 3        |
| H2-b3 l/b3    | 5/1/4  | 1.00       | 0.16 | 0.33 | 0.8438 | 1.0000 | 5        |
| H2-c l/c      | 9/1/0  | 10.00      | 0.58 | 1.25 | 0.0039 | 0.6992 | 3        |
| H2-c2 l/c2    | 10/0/0 | 13.00      | 0.58 | 2.26 | 0.0020 | 0.3711 | 4        |

## H1

Original hypothesis: Sol 6.1 low dominates Sonnet medium on all three axes. Means show +1.30 quality points and lower cost, but longer time. The asserted joint dominance is not observed; quality equivalence or superiority is not established.

## H2

Luna is faster on 10/10 tasks against a, b, b2 and b3, on 6/10 against c and 9/10 against c2. Quality depends on the comparator: its mean exceeds older setups but is below IQ3\_S. Without a preregistered equivalence margin, equal quality cannot be confirmed. Negligible cost had no numerical threshold.

# Effort, variants and cross-camp comparisons

Read A / B: quality wins, ties, losses; median  $\Delta Q$ ; median A/B ratios. Ratio  $<1$  means fewer resources.

| Question: A/B        | W/T/L | $\Delta Q$ | Time | Cost | p Q    | Holm Q | Dom. /10 |
|----------------------|-------|------------|------|------|--------|--------|----------|
| Sol effort e2/e3     | 5/0/5 | 0.00       | 0.63 | 0.63 | 0.9453 | 1.0000 | 5        |
| Opus effort d2/d     | 4/1/5 | -0.50      | 0.38 | 0.33 | 0.5156 | 1.0000 | 5        |
| Local / hosted b2/e3 | 7/2/1 | 3.00       | 4.76 | 0.29 | 0.0234 | 1.0000 | 0        |
| GLM / Flash f/j      | 3/0/7 | -1.75      | 0.44 | 3.91 | 0.4414 | 1.0000 | 0        |
| IQ3_S / Q2_0 b2/b3   | 8/2/0 | 3.50       | 1.29 | 1.29 | 0.0078 | 1.0000 | 1        |
| Haiku / Sonnet n/g   | 7/0/3 | 1.00       | 2.50 | 0.19 | 0.4160 | 1.0000 | 0        |

## Effort axes

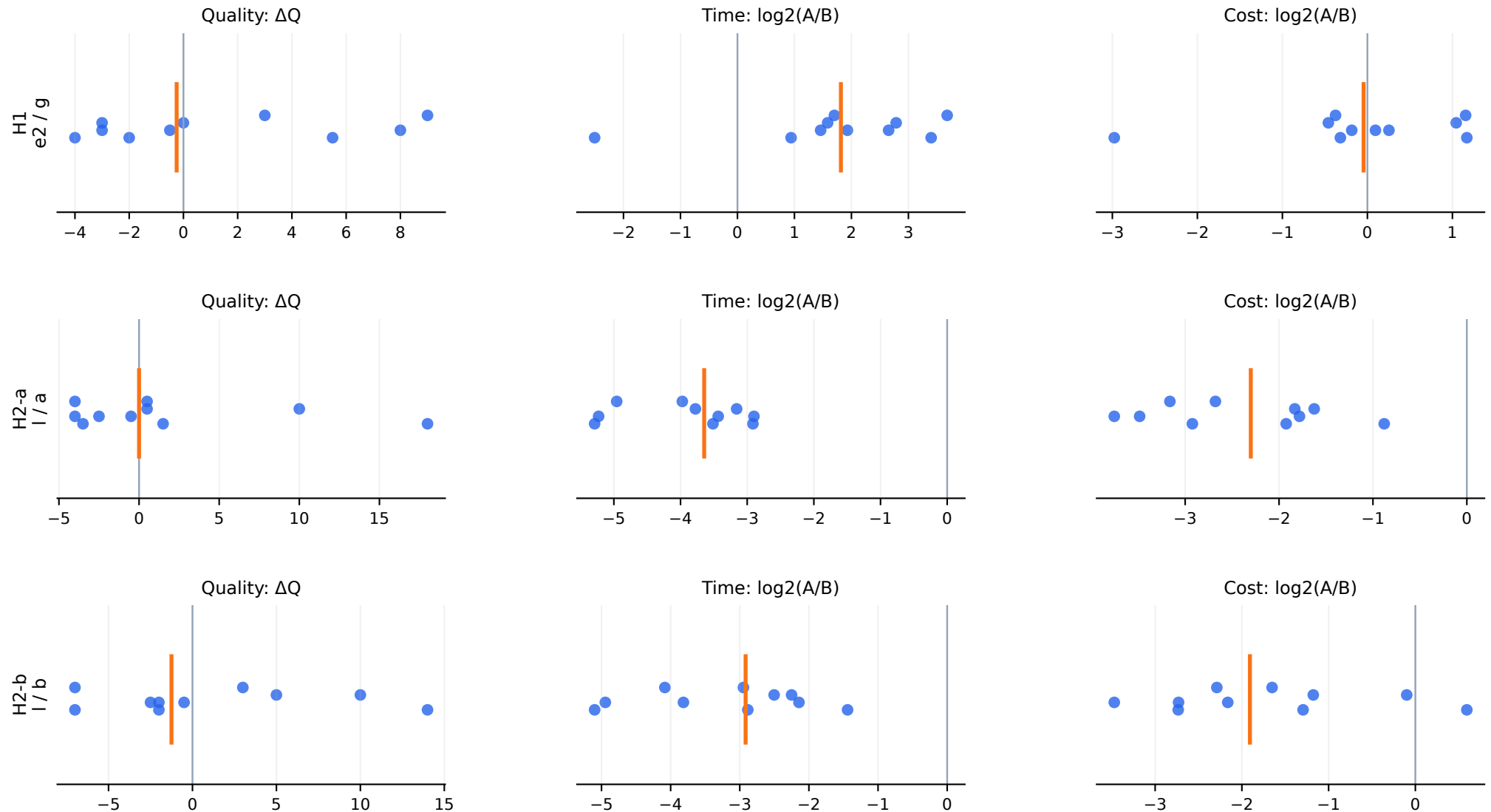
Sol low versus medium: quality wins and losses vary by task; low is faster on 9/10 and cheaper on 10/10. Opus low reduces time and cost, with lower mean quality (21.45 vs 23.70). A nonsignificant test does not justify parity. Cross-cycle differences may also involve panel and environment.

## Interpretation

Dom. /10 counts tasks where A is no worse on all axes and strictly better on at least one; it is not a superiority test. Holm Q here covers all 190 pairs of 20 identities, separately by axis; original DAGs retain their 136-pair family. Neither family yields corrected differences. Other p columns are available in appendix-analysis.json.

# Distributions of paired differences

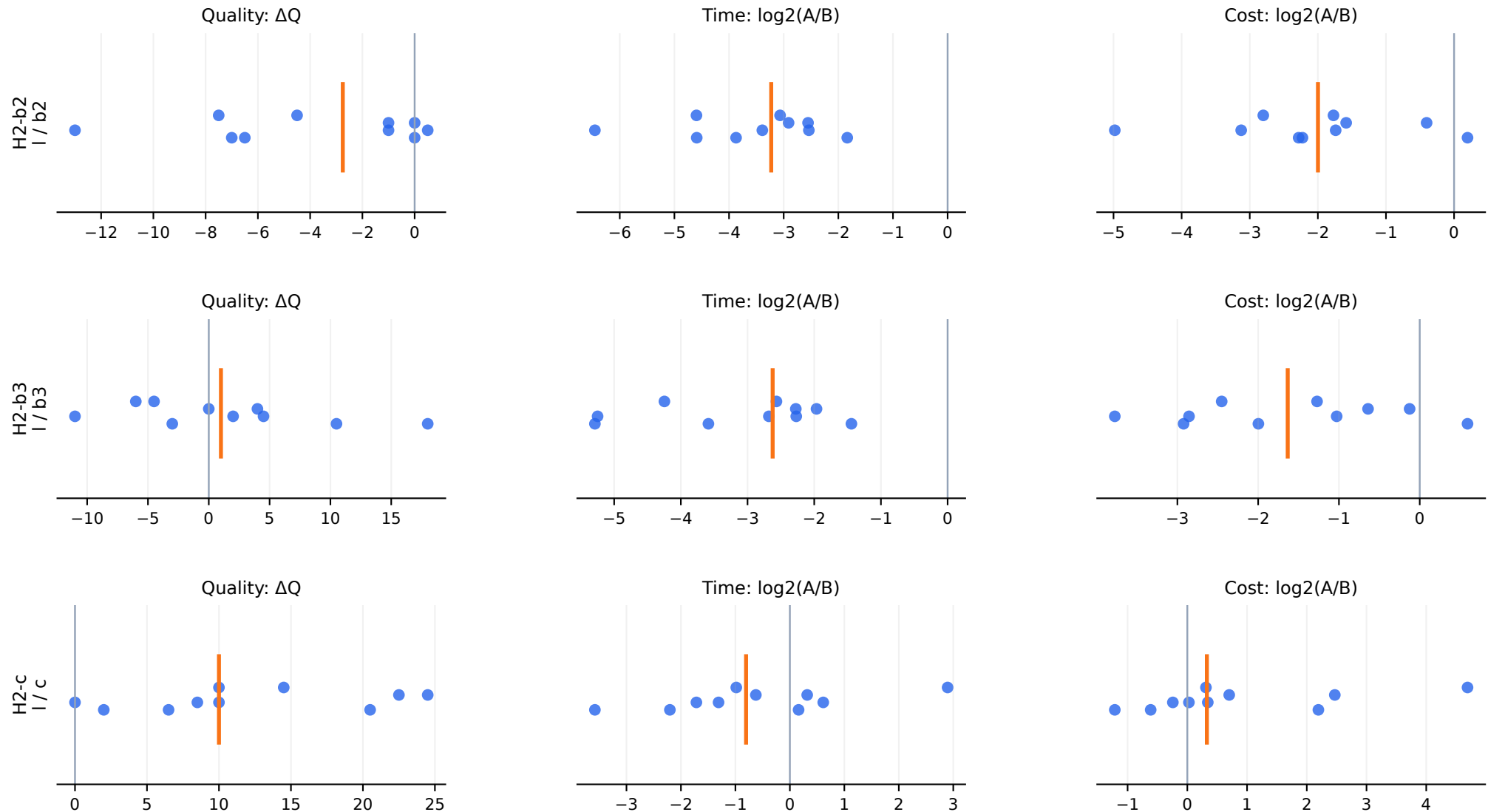
Ten points per comparison.  $\Delta Q = A-B$ ; resources:  $\log_2(A/B)$ . Zero means equal; left means A faster/cheaper.



Orange: median of differences or log-ratios. Per-ticket data: paired-differences.csv. No equivalence interval is estimated.

# Distributions of paired differences

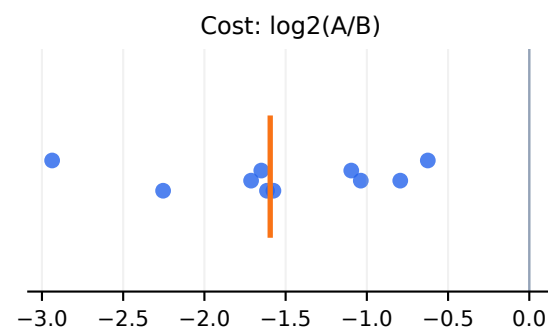
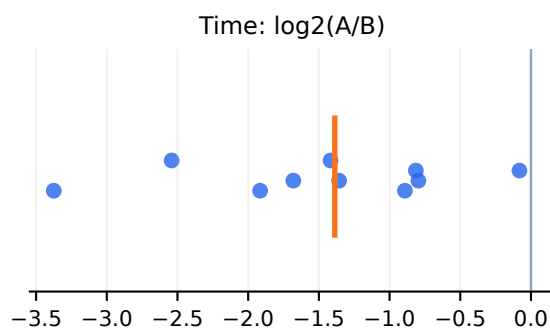
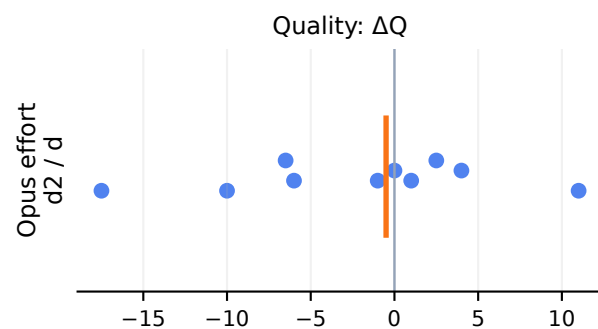
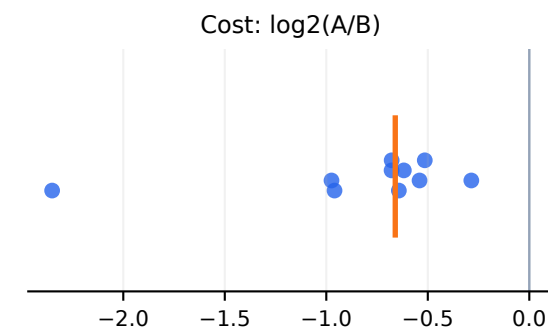
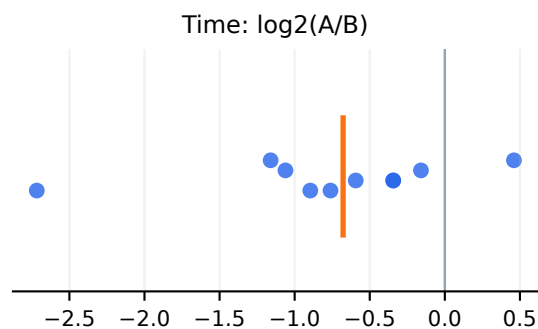
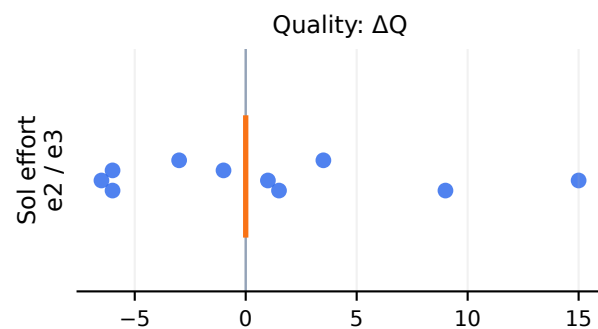
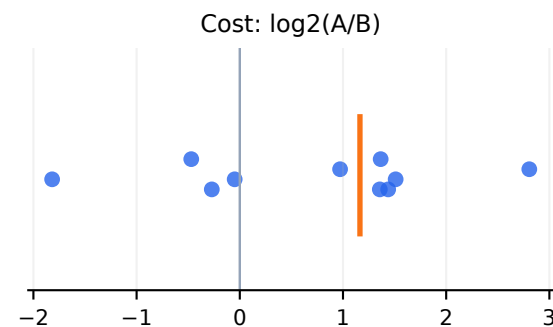
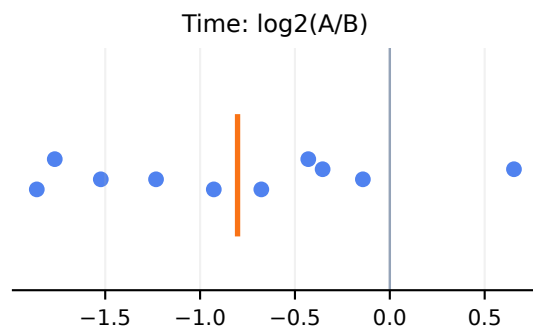
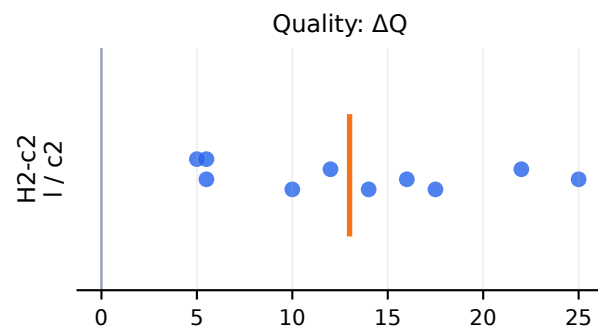
Ten points per comparison.  $\Delta Q = A - B$ ; resources:  $\log_2(A/B)$ . Zero means equal; left means A faster/cheaper.



Orange: median of differences or log-ratios. Per-ticket data: paired-differences.csv. No equivalence interval is estimated.

# Distributions of paired differences

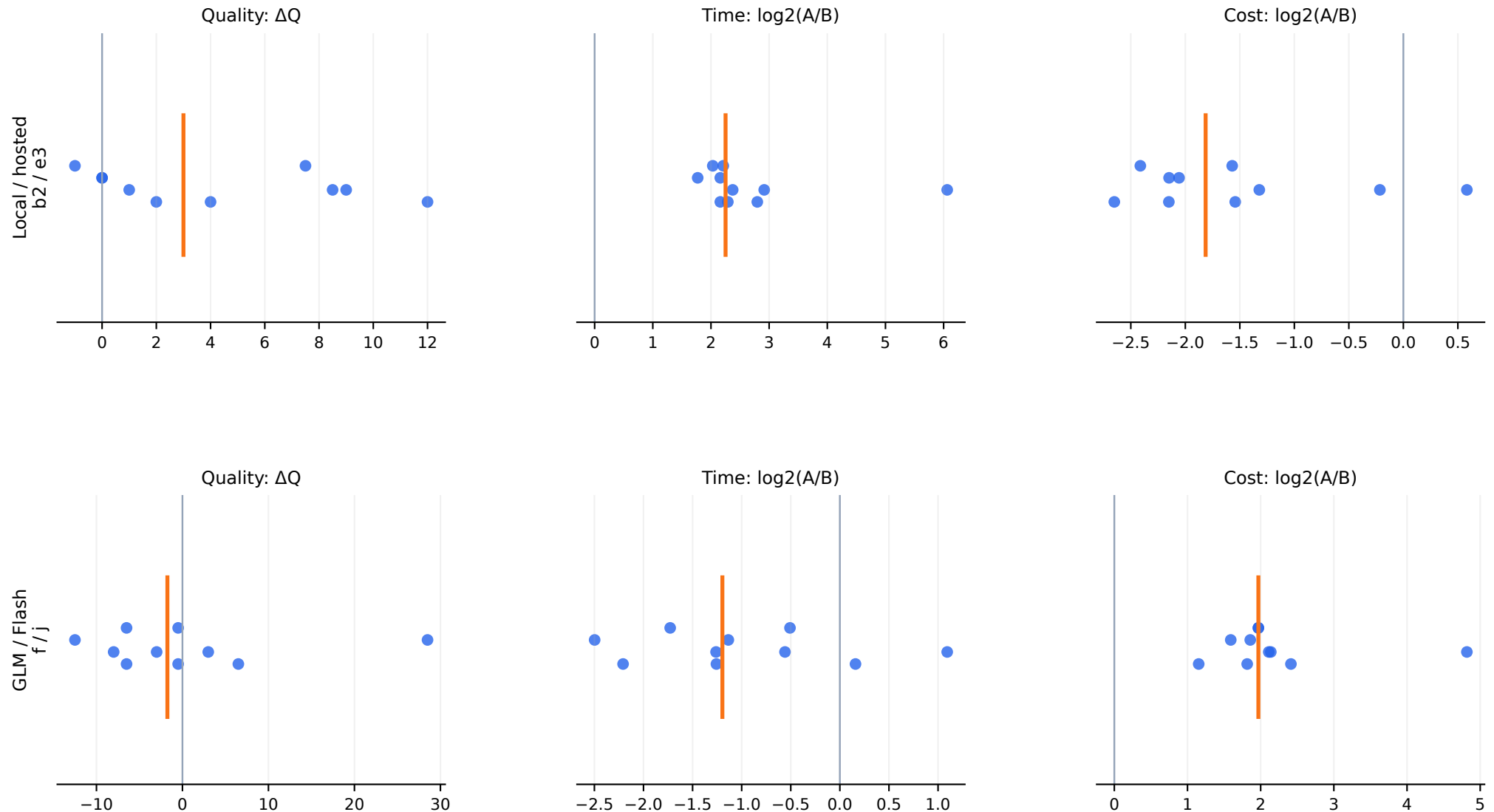
Ten points per comparison.  $\Delta Q = A-B$ ; resources:  $\log_2(A/B)$ . Zero means equal; left means A faster/cheaper.



Orange: median of differences or log-ratios. Per-ticket data: paired-differences.csv. No equivalence interval is estimated.

# Distributions of paired differences

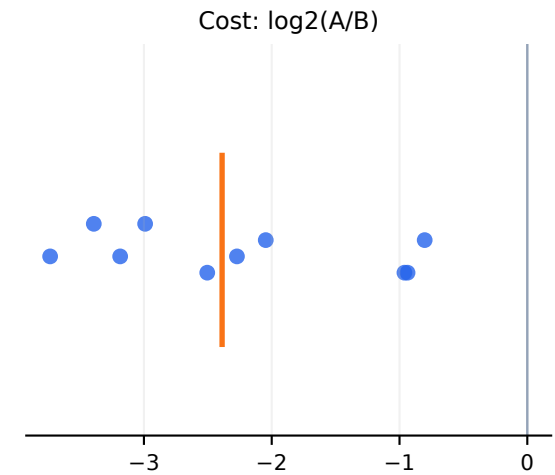
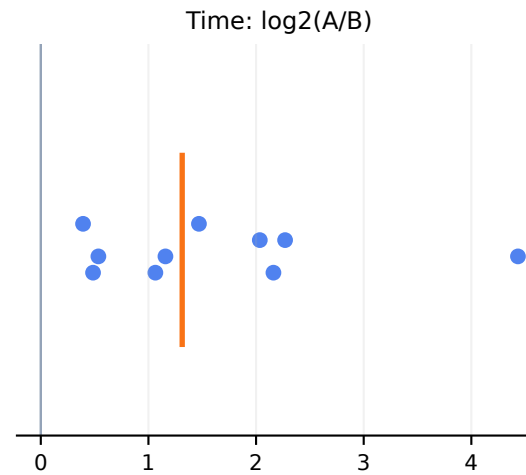
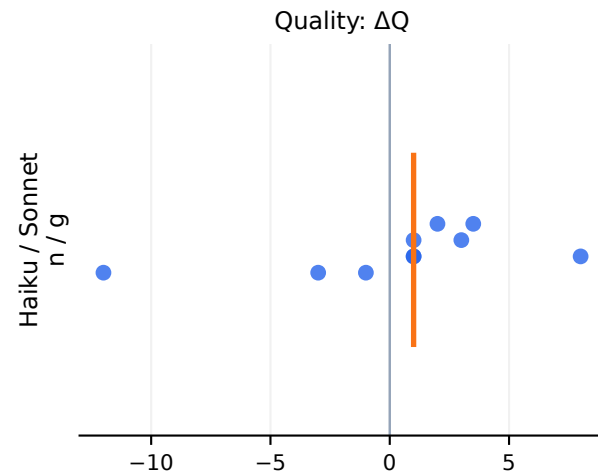
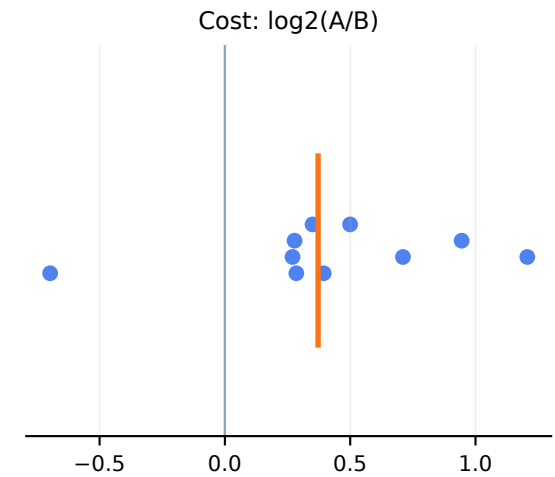
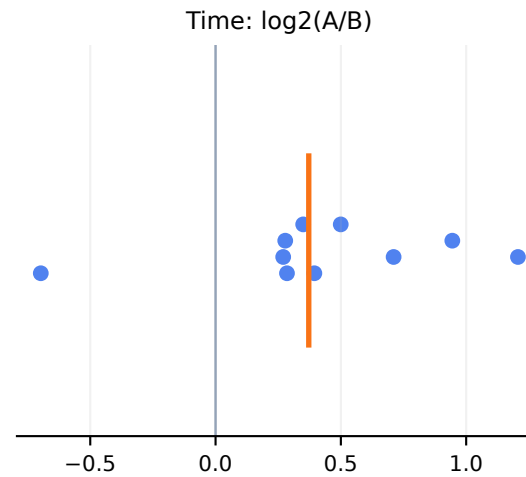
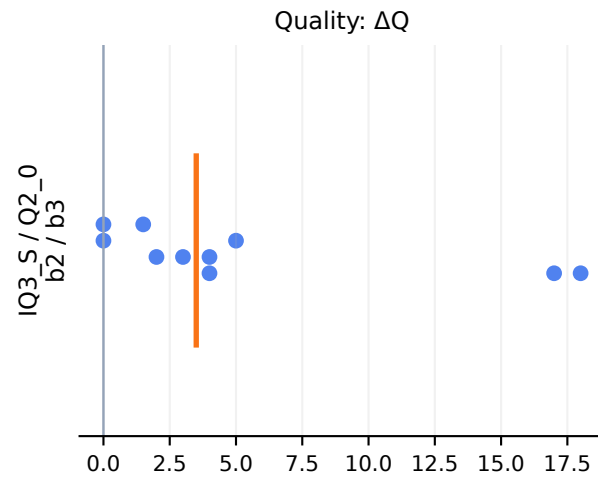
Ten points per comparison.  $\Delta Q = A-B$ ; resources:  $\log_2(A/B)$ . Zero means equal; left means A faster/cheaper.



Orange: median of differences or log-ratios. Per-ticket data: paired-differences.csv. No equivalence interval is estimated.

# Distributions of paired differences

Ten points per comparison.  $\Delta Q = A - B$ ; resources:  $\log_2(A/B)$ . Zero means equal; left means A faster/cheaper.



Orange: median of differences or log-ratios. Per-ticket data: paired-differences.csv. No equivalence interval is estimated.

# Routing classes: a critical reading of the grid

Groups from skills/route/grid.json; operational categories, not fitted statistical clusters.

| Historical class | Arms          | Mean Q /30    | Mean minutes  | Mean USD      |
|------------------|---------------|---------------|---------------|---------------|
| ASSURANCE        | d             | 23.70 - 23.70 | 9.07 - 9.07   | 1.558 - 1.558 |
| MILIEU SOLIDE    | e2, e3, g, f  | 21.35 - 22.65 | 2.09 - 10.58  | 0.205 - 1.286 |
| ECONOMIQUES      | l, n, i, j, k | 21.25 - 22.20 | 5.02 - 60.06  | 0.039 - 0.305 |
| LOCAL-PRIVE      | b2, b3, a     | 20.60 - 26.20 | 39.73 - 63.87 | 0.099 - 0.159 |
| BANNIS           | c, c2         | 8.95 - 10.30  | 9.15 - 10.21  | 0.023 - 0.025 |

## What these groups do not establish

Assurance does not guarantee reliability; solid-middle and economical groups overlap in mean quality. Local setups are not homogeneous: generation, quantization and effort differ. The banned label is an author routing posture supported by low scores here, not a general ban.

## Unassigned identities and stale notices

b, d2 and e are not assigned to these groups; mL/mA are two accounting scopes of one Mistral extension, not two models. j/k notes still say replay underway, while the current snapshot is final with one retained failure each. The report uses the snapshot without silently changing routing policy.

# The 19 historical outcomes, in three classes

141/160 OK and 19 non-OK before the b2 replay; 142/160 OK and 18 non-OK after it. Attempts are not final matrix outcomes.

| Initial class | Count | Identifiers                     | Diagnosis and treatment                                   |
|---------------|-------|---------------------------------|-----------------------------------------------------------|
| DNF           | 4     | 0874-a, 0874-b3, 0333-j, 0211-k | 7200 s cap; all four failures remain zero.                |
| VOID-EMPTY    | 1     | 0333-b2                         | Requested pause to free GPU; archived; replay 30/30.      |
| VOID-EMPTY    | 1     | 0470-c2                         | No valid deliverable; zero retained.                      |
| VOID-EMPTY    | 4     | 0233-j, 0452-j, 0874-j, 1372-j  | OpenRouter quota 403 errors; model attribution withdrawn. |
| VOID-EMPTY    | 3     | 0452-k, 0673-k, 1372-k          | Same quota correction; replays in current series.         |
| DNF-PROVIDER  | 6     | h (SpaceBunny)                  | Endpoint 404, provider unavailable; series excluded.      |

## Attribution corrections

Of nine historical VOID-EMPTY outcomes, one is an author interruption, seven are quota incidents and one is a retained non-delivery. An initial verdict class is not a cause. The six individual h identifiers are absent from the public sources inspected; only the documented count is reported.

## Current state

The five retained failures in current series are the four timeouts and 0470-c2. Historical 19 is not the failure count in the current report. Mistral and Haiku are distinct extensions of that historical matrix. Source: October 6 journal and its two corrections, plus snapshot.json.

# Situation guide: constraints before ranking

Reviewed by a delegated Fable reviewer at the author's request. This guide is neither data-transfer authorization nor an applied routing policy.

| Situation                         | Proposed choice and limit                                                                                                                                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uncleared content / offline       | Local required in this case; IQ3_S if time permits, otherwise defer or reduce the task. Scores do not override confidentiality.                                                                                                                             |
| Interactive, short wait preferred | Sonnet: 2.09 min mean (1.13 median); Opus low: 3.03 (2.82); Haiku: 6.11 (3.03); Luna: 5.02 (3.84). Observed values, not guaranteed times; Sonnet/Opus low is not separated even nominally ( $p = 0.16$ ).                                                   |
| Bulk work, small API budget       | Luna: USD 0.0388 mean, 0.0278 median per ticket. Haiku: 0.1257 mean, 0.0228 median; the mean carries the tail risk from long tasks and the 5x rate above 100k prompt tokens. Paired cost difference not even nominal ( $p = 0.36$ , Haiku cheaper on 5/10). |
| Asynchronous, GPU available       | IQ3_S: 26.20/30, 46.11 min. Reserve a GPU window; the test does not measure multiuser throughput or another machine. Best local mean; gaps to other local arms are nominal only.                                                                            |
| GPU busy                          | If content is cleared, pick from the latency or budget rows; otherwise wait. The 0333-b2 pause was an author-requested GPU release (replay 30/30), not a model failure.                                                                                     |
| Retries matter                    | Budget every attempt and the judging, not just the last success. Mistral mA covers 26 of 35 preserved attempts, not the campaign bill. For Haiku, judging (USD 1.48) cost more than the candidate runs (1.26).                                              |

Haiku/Sonnet example: with linear delay valuation, mean total costs cross at about USD 2.66 per hour of delay per ticket (the report's erg·h). Above this, Sonnet has lower valued total cost; below it, Haiku. This ignores the quality difference (nominal only,  $p = 0.42$ ) and does not state an author preference.

# Guide: off-peak, quotas and availability

Pricing checked October 9, 2026; guide rates are separate from frozen historical tournament costs.

## Direct DeepSeek: schedule flexible work

Official documentation gives off-peak rates at half of peak. Peaks Monday-Friday: 01:00-04:00 and 06:00-10:00 UTC, excluding Chinese public holidays; other hours, weekends and Chinese public holidays are off-peak. Use UTC rather than a fixed Paris hour that changes with DST. Flash cache-miss input: USD 0.15/0.30 per million; output: 0.60/1.20; cache hit: 0.003/0.006. Reseller pricing need not match.

## Measurement and accounting

Per-request events are not reaudited here. Task finish time can be an imperfect proxy when a task crosses a rate boundary. This guide does not retrospectively reprice tournament points; compare provider billing with client estimates for a new campaign.

## Quotas and caps: protect the judges too

The journal records depleted OpenAI credits, an OpenRouter monthly cap raised from USD 40 to 60, and a withdrawn free endpoint. The OpenRouter cap stalled the entire panel, whose three judges used that provider. Before a new campaign: check credits, caps, concurrency limits (DeepSeek: Flash 2,500, Pro 500) and judge budget; reserve headroom; preserve incident evidence and distinguish provider and model errors.

Official DeepSeek source (accessed October 9): [api-docs.deepseek.com/quick\\_start/pricing/](https://api-docs.deepseek.com/quick_start/pricing/)

# Provenance and ticket 1046 exit criteria

Supplements computed from ~/arena (padme); the public snapshot gives the same numbers. No private task content is included.

## Reproduction

The original generator calls tournament-report-completion.py to append pages 13-27 in both languages. Distributions and tests are recomputed from the snapshot built from ~/arena (snapshot.json); appendix-analysis.json retains all 190 pairs and paired-differences.csv the 130 observations of thirteen displayed comparisons. Base numeric data and the DAG 136 tests per axis are unchanged.

## Sources and traceability limits

Frozen public snapshot e352f74...: retained results, attempt audits and Mistral/Haiku costs. Grid: skills/route/grid.json. Hypotheses: ticket 1024, October 5 at 08:40Z. Outcomes: October 6 journal and corrections. On October 9 the generator rerun in --arena mode on the private archives (padme) reproduced tests.json, comparisons.csv and every series value identically; the public snapshot only adds Haiku 5.5 accounting metadata. Random selection and original invoices are not reaudited. Six h incidents are reported as a count without invented identifiers.

## Exit criteria status

Requested content is now present: plots, grid, H1/H2 verdicts, effort, cross-camp comparisons, distributions, classes, outcomes and guide. Public reproduction and layout are checked. The guide was reviewed by a delegated Fable reviewer on the author's instruction, and its corrections applied. This report is the paired-analysis write-up required by parent 1024. Original task content is not exported.